University of Trento

Computational Physics

Master Degree in Physics

Computational Physics Final Project

A Physics-Informed Neural Network Approach to the Gross-Pitaevskii Equation in General 1D and 2D Trapping Potentials

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Introduction

Neural networks (NNs) are a class of algorithms known as universal function approximators. In recent years, they have been successfully applied to physics problems in the form of Physics-Informed Neural Networks (PINNs), where the NN is trained to approximate the solution of a physical problem, and the training loss function is modified to include physical constraints, such as the governing equation of the system.

In this report, we explore this computational approach by solving problems of increasing complexity, starting from the one-dimensional (1D) stationary Gross-Pitaevskii equation (GPE).

We then move to two-dimensional (2D) problems, where we find a complex-valued solution to a nonlinear Schrödinger equation, reproducing the results of Example 3.1.1 in Raissi, Perdikaris, and Karniadakis 2019. Finally, we solve the stationary GPE in 2D for various potentials and investigate the existence of vortex solutions in the absence of an external potential.

The NN architecture is implemented using the `PyTorch` library, and the optimization is performed using the `L-BFGS` algorithm. The number of layers and neurons is chosen based on the complexity of the problem, but it varies between 2–4 layers and 4–32 neurons per layer. Hyperbolic tangent activation functions are used for all the problems.

The code is available in the GitHub repository `https://github.com/F-Depi/PINN-GPE`

1 1D Gross-Pitaevskii equation With Harmonic Potential

If we take the Gross-Pitaevskii equation in 3D, express the energy in units of $\hbar\omega$, the length in units of $a_{\text{ho}} = \sqrt{\hbar/m\omega}$, we get

$$-\frac{1}{2}\nabla^2\psi(\mathbf{r}) + V_{\text{ext}}(\mathbf{r})\psi(\mathbf{r}) + g|\psi(\mathbf{r})|^2\psi(\mathbf{r}) = \mu\psi(\mathbf{r}) \tag{1}$$

where $g = 4\pi a$ is the interaction strength (positive for repulsive interactions, negative for attractive ones) and μ is the chemical potential. If we look for the ground state solution with a harmonic potential $V_{\text{ext}}(\mathbf{r}) = \frac{1}{2}r^2$ we can write the wave function as

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{4\pi}}\frac{\phi(r)}{r}\,.$$

Normalizing $\psi(\mathbf{r})$ to 1 $\left(\psi(\mathbf{r}) \to \sqrt{N}\psi(\mathbf{r})\right)$ with N the number of particles of the system, equation (1) reduces to

$$-\frac{1}{2}\frac{\mathrm{d}^2\phi(r)}{\mathrm{d}r^2} + \frac{1}{2}r^2\phi(r) + Na\left(\frac{\phi(r)}{r}\right)^2\phi(r) = \mu\phi(r).$$

where $\phi(x)$ is the radial part of the wave-function that will be approximated by our NN, Na can be treated as one parameter that controls the strength of the interaction (we explore $Na = 1, 10, 100$) and μ is the chemical potential of the system, which is the eigenvalue of the problem and is related to the energy of the system. Because we don't know μ a priori, we need to include it as a trainable parameter in the NN. [1] From physical considerations we can fix the boundary conditions $\phi(r = \epsilon) = 0$ and $\phi(r = R) = 0$, where $\epsilon = 10^{-7}$ is chosen to avoid the singularity at $r = 0$ and $R = 6$ is a sufficiently large radius to capture the behavior of the wave-function.

The loss function is defined as

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{norm}} \\
\mathcal{L}_{\text{PDE}} &= \frac{1}{N_f}\sum_{i=1}^{N_f}\left(-\frac{1}{2}\frac{\mathrm{d}^2\phi(r_i)}{\mathrm{d}r^2} + \frac{1}{2}r_i^2\phi(r_i) + Na\left(\frac{\phi(r_i)}{r_i}\right)^2\phi(r_i) - \mu\phi(r_i)\right)^2 \\
\mathcal{L}_{\text{BC}} &= \phi(\epsilon)^2 + \phi(R)^2 \\
\mathcal{L}_{\text{norm}} &= \left(\frac{R-\epsilon}{N}\sum_{i=1}^{N}\phi(r_i)^2 - 1\right)^2.
\end{aligned} \tag{2}$$

[1] In this problem we are looking for the ground state solution, so we should be minimizing μ during the training. However, luckily for us, this was not necessary, and the optimizer always converged to the lowest eigenvalue. We will not be this lucky going forward.

Enforcing the normalization with $\mathcal{L}_{\text{norm}}$ is needed as $\phi(r) \equiv 0$ would have been a valid solution otherwise. We chose $N_f = 1000$ distributed with Latin Hypercube Sampling (LHS) in the interval $[\epsilon, R]$ and use `L-BFGS` as optimizer. Given the great chance of the gradient to get stuck in a local minimum, we stop the optimization after 100 iterations with the same loss, [2] and repeat the same optimization 5 times, keeping the best result. We find this method to be more effective and efficient than using a stochastic optimizer such as `Adam` to initialize the weights of the network.

Figures 1 and 2 show the results for a small NN with 4 neurons per layer and 1 hidden layer (N4L1) for $Na = 1$ and $Na = 10$ respectively.

As we can see, even with a model of $(1 \times 4 + 4 \times 1) + (4 + 1) = 13$ trainable parameters we can achieve a good agreement with the results obtained in a previous report, [3] with a relative $\mathbb{L}_2$ error of 4.67×10^{-4} for $Na = 1$ and 9.55×10^{-5} for $Na = 10$. Considering that the `numerov` data was obtained with a grid resolution of 0.001 the agreement can be considered good.

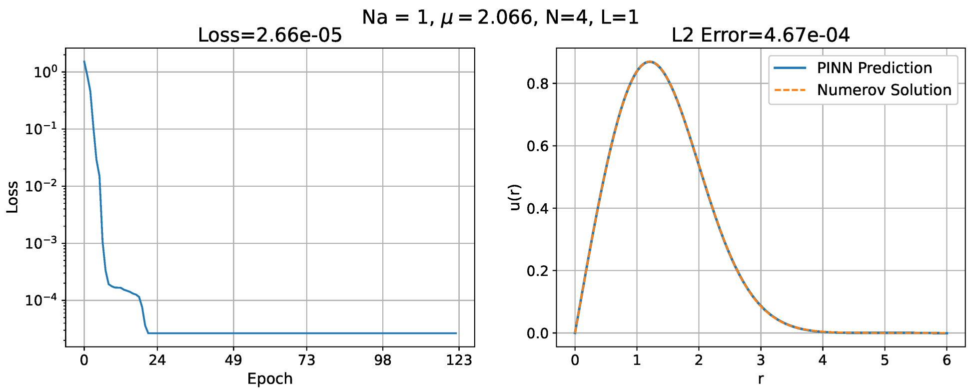

Figure 1: Results for the 1D GPE with $Na = 1$ using a NN with 1 hidden layer and 4 neurons per layer (N4L1). The final loss was 2.66×10^{-5}, the relative $\mathbb{L}_2$ is 4.67×10^{-4} computed with respect to the `Numerov` solution obtained from the previous reports. `L-BFGS` optimizer was stopped after 100 iterations with the same loss.

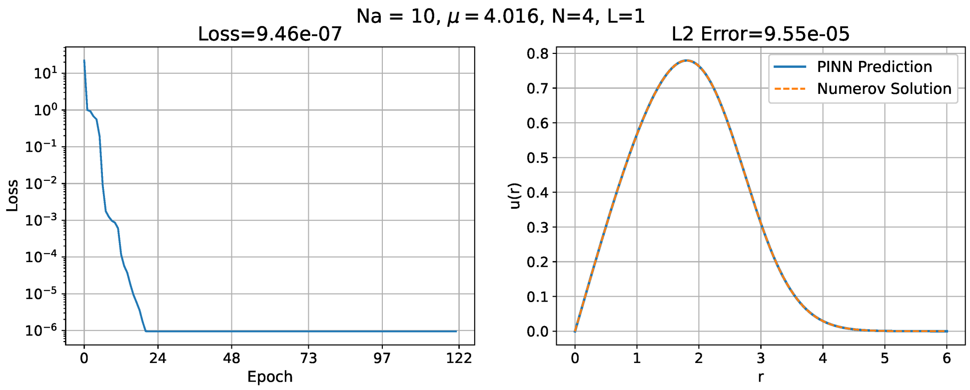

Figure 2: Results for the 1D GPE with $Na = 10$ using a NN with N4L1. The final loss was 9.46×10^{-7}, the relative $\mathbb{L}_2$ error is 9.55×10^{-5} computed with respect to the `Numerov` solution obtained from the previous reports. `L-BFGS` optimizer was stopped after 100 iterations with the same loss.

More complex architectures with more layers and neurons were tested, but they didn't show any significant improvement in the results, on the other hand even smaller NNs gave worse results. The losses and $\mathbb{L}_2$ errors for a variety of numbers of neurons and layers are reported in Table 1 for the

[2] We define *same loss* as $|\mathcal{L}_i - \mathcal{L}_{i-1}|/\mathcal{L}_i < 10^{-6}$, where i is the iteration number.

[3] The data we are referring to is obtained with the `numerov` method, for the Homework 3 – Mean Field Methods F. De Paoli and F. Giuliani

$Na = 0$ case, of which we know the analytical solution

$$\phi(r) = \frac{2}{\sqrt[4]{\pi}} r e^{-r^2/2}, \quad \mu = \frac{3}{2}.$$

Table 2 shows instead the effect of changing the number of sample points used in the training of the NN, always for $Na = 0$, for a sufficiently complex architecture with N16L3. After 1000 points the improvement in the $\mathbb{L}_2$ error is not significant, but we can positively correlate a low loss with a low $\mathbb{L}_2$ error, as expected.

Neurons per Layer	**Hidden Layers**	**Loss**	$\mathbb{L}_2$ **Error**
2	1	7.22×10^{-4}	4.05×10^{-3}
	2	7.95×10^{-5}	8.05×10^{-4}
	3	3.64×10^{-5}	5.46×10^{-4}
	4	6.34×10^{-5}	8.40×10^{-4}
3	1	1.65×10^{-4}	1.52×10^{-3}
	2	5.77×10^{-6}	1.81×10^{-4}
	3	8.90×10^{-6}	2.43×10^{-4}
	4	4.19×10^{-6}	6.67×10^{-5}
4	1	2.28×10^{-6}	7.95×10^{-5}
	2	3.40×10^{-6}	1.18×10^{-4}
	3	4.73×10^{-6}	1.71×10^{-4}
	4	2.49×10^{-6}	1.30×10^{-4}
8	1	5.98×10^{-6}	1.87×10^{-4}
	2	2.76×10^{-6}	1.12×10^{-4}
	3	3.14×10^{-7}	4.81×10^{-5}
	4	3.06×10^{-6}	1.26×10^{-4}
16	1	5.84×10^{-6}	1.89×10^{-4}
	2	8.90×10^{-7}	6.23×10^{-5}
	3	1.78×10^{-6}	1.04×10^{-4}
	4	2.69×10^{-6}	8.75×10^{-5}

Table 1: Loss and relative $\mathbb{L}_2$ error for different NN architectures for the 1D GPE with $Na = 0$ (analytical solution available). Results are consistently good across all architectures from 4 neurons per layer, with no clear improvement observed for more complex architectures.

Sample Points (N_f)	**Loss**	$\mathbb{L}_2$ **Error**
10	6.84×10^{-7}	1.97×10^{-2}
100	1.53×10^{-6}	7.05×10^{-4}
500	2.40×10^{-6}	1.48×10^{-4}
1000	8.34×10^{-7}	4.02×10^{-5}
5000	3.55×10^{-6}	8.42×10^{-5}
10000	4.05×10^{-6}	1.37×10^{-4}
20000	9.76×10^{-7}	5.08×10^{-5}

Table 2: Loss and relative $\mathbb{L}_2$ error for different numbers of sample points for the 1D GPE with $Na = 0$, using a sufficiently expressive NN with N16L3. The $\mathbb{L}_2$ error decreases significantly as the number of points increases from 10 to 1000, but does not show a clear improvement beyond that. Low loss is generally correlated with low $\mathbb{L}_2$ error, as expected.

Finally, considering the troubles we went through to attempt to solve the GPE with $Na = 100$ (very strong interaction) with both the `numerov` and the `variational` method, we were curious to see how the NN would perform in this case. A first run with N4L1 gave only a 3.46×10^{-4} loss, but it was sufficient to increase to N8L2 to get a much better loss of 1.01×10^{-6}. In the previous report the solutions obtained from `numerov` and the `variational` method were always in good agreement, except for the $Na = 100$ case. It was in fact not clear which of the two methods was giving the correct solution, as both the procedures showed some flaws in the convergence of the parameters, but the PINN training was straightforward, and the result is clearly in good agreement with the `variational` data with a relative $\mathbb{L}_2$ error of 8.99×10^{-5} (Figure 3).

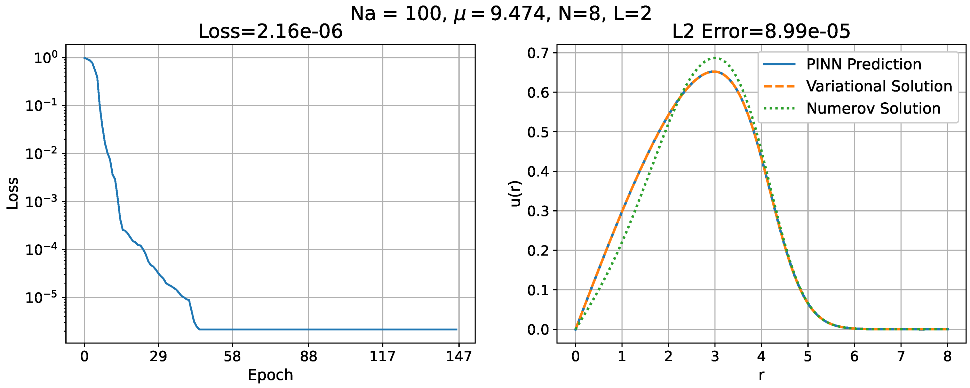

Figure 3: Results for the 1D GPE with $Na = 100$ using a NN with N8L2. The relative $\mathbb{L}_2$ is 8.99×10^{-5} computed with respect to the data obtained from the previous reports with the `variational` method. `L-BFGS` optimizer was stopped after 100 iterations with the same loss.

2 Time-dependent Gross-Pitaevskii equation

We can now move to a 2D problem by replicating the example 3.1.1 in Raissi, Perdikaris, and Karniadakis 2019, where the authors solve a nonlinear Schrödinger equation with boundary conditions

$$\begin{aligned}
&ih_t + 0.5h_{xx} + |h|^2 h = 0, \quad x \in [-5, 5], \quad t \in [0, \pi/2],\\
&h(0, x) = 2\ \mathrm{sech}(x),\\
&h(t, -5) = h(t, 5),\\
&h_x(t, -5) = h_x(t, 5),
\end{aligned}$$

where the subscripts denote partial derivatives. This corresponds to the evolution of a soliton solution with unitary interaction and no potential under the time-dependent GPE.

In this case the solution is complex, so we need to separate real and imaginary parts $h(t, x) = u(t, x) + iv(t, x)$, such that

$$\mathcal{N}(t, x) = [u(t, x),\ v(t, x)] \tag{3}$$

where $\mathcal{N}$ is a multi-output NN with 2 inputs and 2 outputs.

The loss function is defined as

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_{\mathrm{PDE}} + \mathcal{L}_{\mathrm{IC}} + \mathcal{L}_{\mathrm{BC}}\\
\mathcal{L}_{\mathrm{PDE}} &= \frac{1}{N_f}\sum_{i=1}^{N_f}\left|ih_t(t_i, x_i) + 0.5h_{xx}(t_i, x_i) + |h(t_i, x_i)|^2 h(t_i, x_i)\right|^2\\
\mathcal{L}_{\mathrm{IC}} &= \frac{1}{N_0}\sum_{i=1}^{N_0}|h(0, x_i) - 2\ \mathrm{sech}(x_i)|^2\\
\mathcal{L}_{\mathrm{BC}} &= \frac{1}{N_b}\sum_{i=1}^{N_b}|h(t_i, -5) - h(t_i, 5)|^2 + |h_x(t_i, -5) - h_x(t_i, 5)|^2
\end{aligned}$$

where we now have N_0 points for the initial condition and N_b points to enforce the symmetry of the solution at the boundaries. Given that we have a non-zero initial condition there is no need to enforce the normalization this time. We use $N_f = 20000$, $N_0 = 50$ and $N_b = 50$ points, all distributed with LHS in the appropriate intervals, as it was done by Raissi, Perdikaris, and Karniadakis 2019.

The increased dimensionality of the problem forces us to use a more complex NN architecture as the first attempts with only 4–8 neurons and 1–2 hidden layers could not go below a loss of 10^{-2}. Eventually a first satisfactory result was obtained with N16L3, with a loss of 1.28×10^{-4}. Figure 4 shows the results for an even bigger NN with N32L3, with a loss of 1.86×10^{-5}.

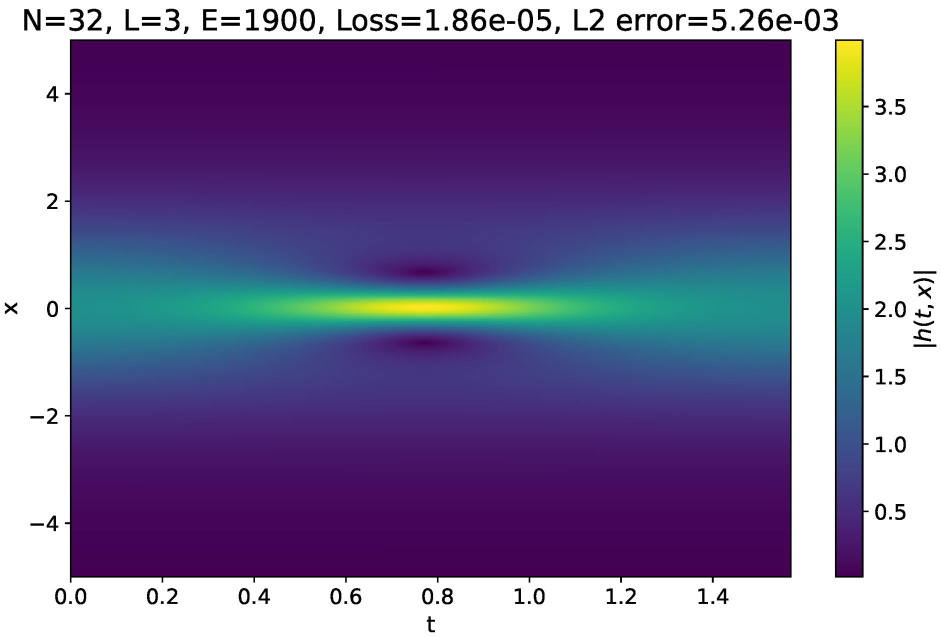

Figure 4: Results for the time-dependent GPE with a NN with N32L3. The relative $\mathbb{L}_2$ is 5.26×10^{-3} computed with respect to the analytical solution. `L-BFGS` optimizer was stopped after 100 iterations with the same loss.

Thanks to the data of the exact solution provided in the GitHub repository of Raissi, Perdikaris, and Karniadakis 2019 (`https://github.com/maziarraissi/PINNs`) we can also compute the relative $\mathbb{L}_2$ error of our solution with respect to the analytical one, which gave 1.45×10^{-2} for the first satisfactory result and 5.26×10^{-3} for the bigger NN, showing again that a lower loss correlates with a better solution. Considering that Raissi et al. used a N100L5 NN and obtained a relative $\mathbb{L}_2$ error of 1.97×10^{-3}, we are very satisfied with our results.

Figure 5 shows some slices of the solution for the bigger NN to appreciate the agreement with the analytical solution, even in the highly nonlinear regions of the domain.

3 Ground States of the GPE in Various Trapping Potentials

To further test the reliability of the method we now explore the capability of the NN to solve the GPE with a double well potential.

3.1 1D Double Well

We report here the GPE in 1D

$$-\frac{1}{2}\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \left(V_{\text{ext}}(x) + g|\psi(x)|^2 - \mu\right)\psi(x) = 0$$

where $V_{\text{ext}}(x)$ is a potential of our choice, g is the interaction strength and μ is the chemical potential, which is again a trainable parameter.

We will solve the problem in $x \in [x_l, x_r]$ depending on the potential and asking for $\psi(x_l) = \psi(x_r) = 0$ as boundary conditions, which are physically reasonable.

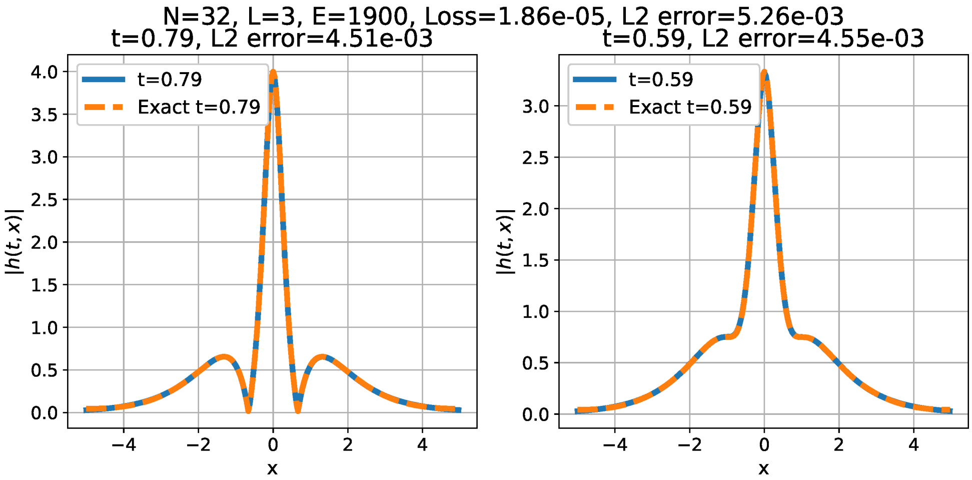

Figure 5: Slices of the results for the time-dependent GPE with N32L3. We can appreciate the NN ability to capture the solution even in the highly nonlinear regions of the domain.

Starting from a symmetric double well potential

$$V_{\text{ext}}(x) = \left(x^2 - a\right)^2 \tag{4}$$

with $a = 1, 2$ so that the middle barrier is respectively $V(0) = 1$ and $V(0) = 4$. We can use $\mathcal{L}_{\text{PDE}}$ with 1000 sample points, $\mathcal{L}_{\text{BC}}$ and $\mathcal{L}_{\text{norm}}$ losses like the ones in (2). The results are shown in Figures 6 and 7 for a NN with N4L2 and $g = 1$.

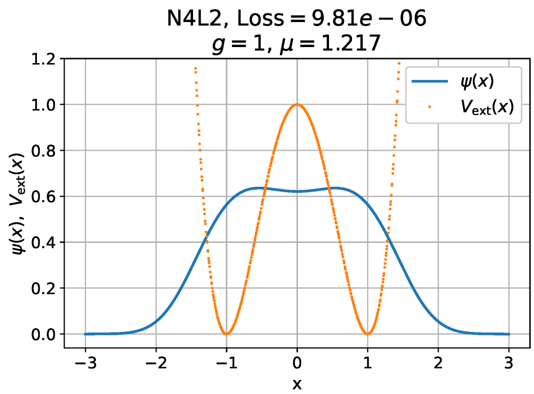

Figure 6: Results for the 1D GPE with the symmetric double well potential from equation (4) with $a = 1$ and $g = 1$ using a NN with N4L2.

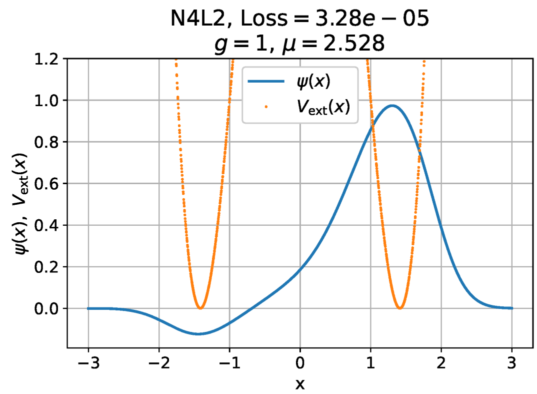

Figure 7: Results for the 1D GPE with the symmetric double well potential from equation (4) with $a = 2$ and $g = 1$ using a NN with N4L2.

We notice immediately that for the case $a = 1$ we found the ground state solution, while for $a = 2$ our wave function has a node, is not symmetric and has a higher μ. This is a clear sign that we found an excited state solution, or not even a physical one.

To help the optimizer find the ground state, we can incorporate a physical constraint that we have so far neglected: the positivity of the wave function. We enforce this by appending a `pytorch.nn.Softplus()` layer at the end of the neural network, which ensures a smooth, strictly positive output. [4] Unfortunately this method alone did not succeed as the NN was still trying to approximate the excited state and the optimizer got stuck in a high loss local minimum. We also tried to initialize the weights of the NN with a gaussian, but the result is always the same (even increasing the NN complexity or the sample points) and is shown in Figure 8.

[4] In practice, enforcing positivity caused the optimizer to become stuck at a loss value of 1, corresponding $\psi(x) \equiv 0$.

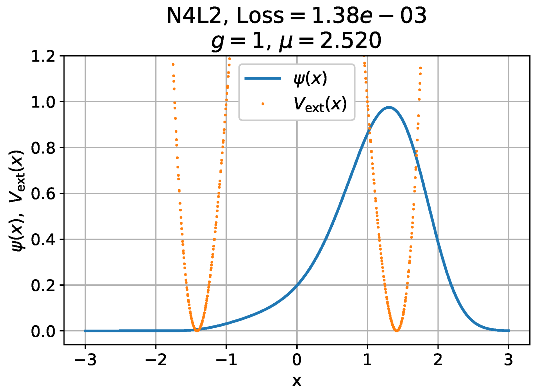

Figure 8: Another failed attempt at finding the ground state of the system under the potential in equation (4), this time we also tried to enforce positivity and used a gaussian initialization. Increasing the NN complexity or the number of sample points had no effect.

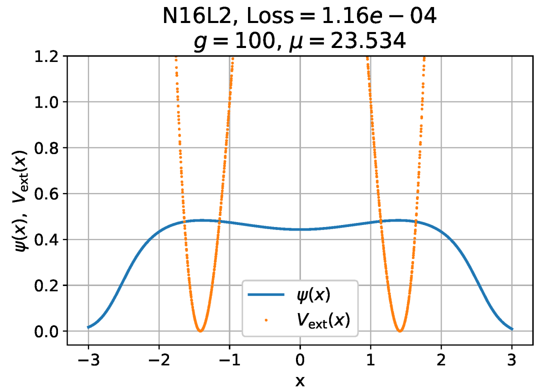

Figure 9: Increasing the strength of the interaction forces the wave function to expand over both the wells. This time we are confident that we found the ground state given the symmetry and the Loss of 2.9×10^{-4}.

What ultimately worked was increasing g to 100, so that the system was highly interacting, allowing the wave function to freely spread over the domain. In this case we found that a N16L2 NN helped, and the results are shown in Figure 9. [5] Now we can start from the NN trained at $g = 100$, and optimize it for $g = 1$, the result is shown in Figure 10. In this case we also find beneficial to increase the number of sample points to 2000.

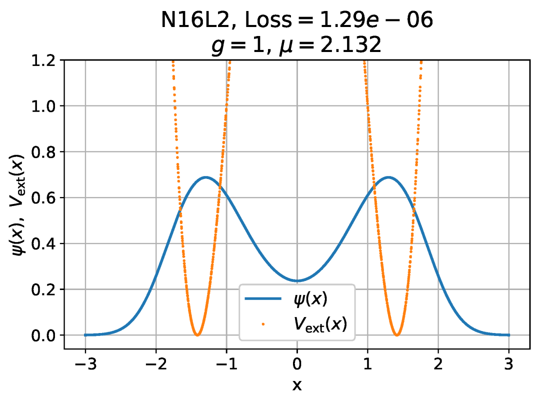

Figure 10: Starting from the NN trained at $g = 100$ in Figure 9 and optimizing it for $g = 1$ we can find the ground state solution for the double well potential with $a = 2$.

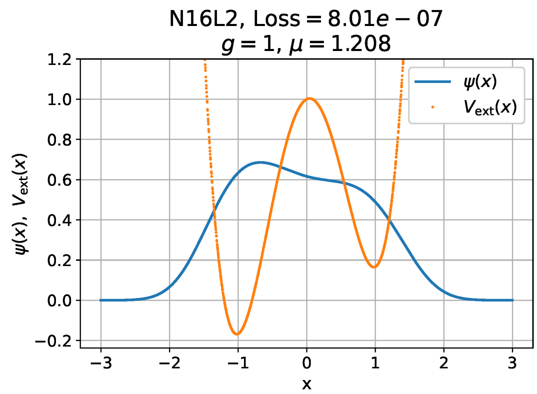

Figure 11: N16L2 positive NN trained for $g = 100$ and then retrained for $g = 1$ with the non-symmetric double well potential from equation (5) with $a = 1$ and $b = 6$. The wave function is more localized in the left well, as expected.

Finally, we can also use a non-symmetric double well potential

$$V_{\text{ext}}(x) = \left(x^2 - a\right)^2 + \frac{x}{b}\,. \tag{5}$$

Using $a = 1$, $b = 6$ we can apply the same method as before by first finding a solution for $g = 100$

To overcome this issue, we strengthened the normalization constraint by increasing its weight, $\mathcal{L}_{\text{norm}} \to 5\,\mathcal{L}_{\text{norm}}$, which encouraged the network to grow away from the trivial solution.

[5] The reader might wonder whether relaxing the normalization to $\int |\psi|^2\,dx = \alpha,\ \alpha > 1$ would have worked too. Indeed, under the rescaling $\psi \mapsto \sqrt{\alpha}\,\psi$ one has $|\psi|^2 \mapsto \alpha|\psi|^2$, which is equivalent to $g \mapsto \alpha g$. Thus, the two approaches are physically identical, and increasing g was simply the more natural knob to turn, that also kept ψ normalized to 1.

and then retraining the NN for $g = 1$. The result is shown in Figure 11, where we can see that the wave function is more localized in the left well, as expected.

These results, obtained by first finding a solution with a high interaction strength, can probably be found by varying the external potential instead and slowly changing it to the final configuration. But given that the potential can have many parameters and that later on we will have to deal with a 2D one, using the single number g to control the evolution of the solution can be more convenient.

A different approach to address the eigenvalue problem under some approximation is reported in Appendix A, where we find multiple eigenstates of the problem at the same time.

3.2 2D Double Well

We now extend the problem to 2D, where the NN takes (x, y) as inputs. The loss is defined as

$$\begin{aligned}
\mathcal{L} &= \mathcal{L}_{\mathrm{PDE}} + \mathcal{L}_{\mathrm{BC}} + 10\,\mathcal{L}_{\mathrm{norm}} \\
\mathcal{L}_{\mathrm{PDE}} &= \frac{1}{N_f}\sum_{i=1}^{N_f}\left(-\frac{1}{2}\frac{\mathrm{d}^2\phi(r_i)}{\mathrm{d}r^2} + \frac{1}{2}r_i^2\phi(r_i) + Na\left(\frac{\phi(r_i)}{r_i}\right)^2\phi(r_i) - \mu\phi(r_i)\right)^2 \\
\mathcal{L}_{\mathrm{BC}} &= \frac{1}{N_b}\sum_{i=1}^{N_b}(\phi(r_i))^2 \\
\mathcal{L}_{\mathrm{norm}} &= \left(\frac{R-\epsilon}{N}\sum_{i=1}^{N}\phi(r_i)^2 - 1\right)^2 .
\end{aligned} \tag{6}$$

For the harmonic potential $V_{\mathrm{ext}}(x, y) = \frac{1}{2}(x^2 + y^2)$ with $g = 1$ the optimizer converges without difficulty, and the results are shown in Figure 12.

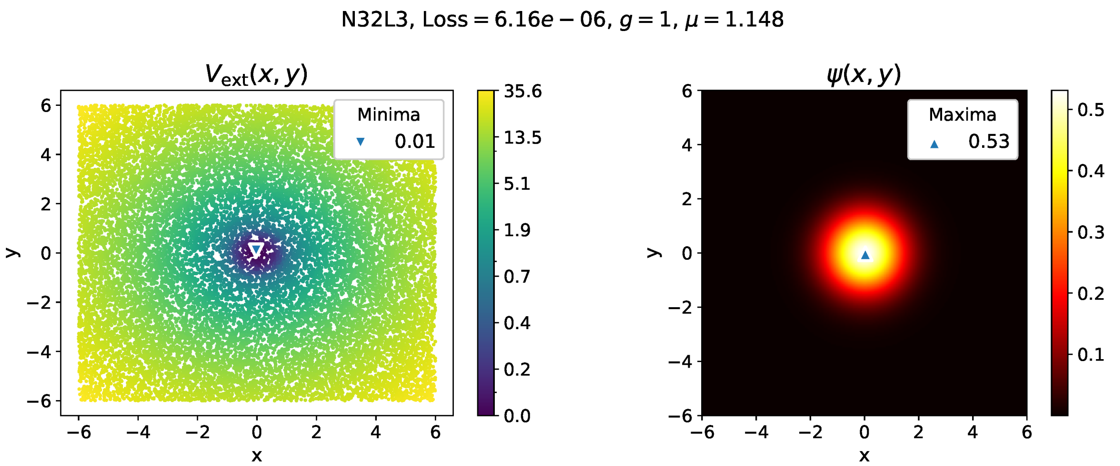

Figure 12: Results for the 2D GPE with a harmonic potential and $g = 1$ using a NN with N32L3. Gaussian initialization was used to help the optimizer.

When trying a double well potential (even a symmetric one) we find ourselves in the same situation encountered in the 2D case, where, even with a gaussian initialization and a positivity constraint on the NN, the optimizer is content to only grow the wave function in one of the wells (no matter which one). We are therefore forced to use the same strategy described at the end of section 3.1: we first find a solution with a high interaction strength, which allows the wave function to spread over both wells, and then retrain the NN for smaller values of g.

We can immediately look at a non-symmetric double well potential

$$V_{\mathrm{ext}}(x, y) = x^4 + \left(y^2 - 2\right)^2 + \frac{y}{6} + xy\,. \tag{7}$$

We first train a N32L3 NN for $g = 40$, the result is shown in Figure 13, and then retrain it for $g = 1$, the result is shown in Figure 14. We can appreciate how the wave function is more localized in the lower well, while still showing a significant density in the higher well as well.

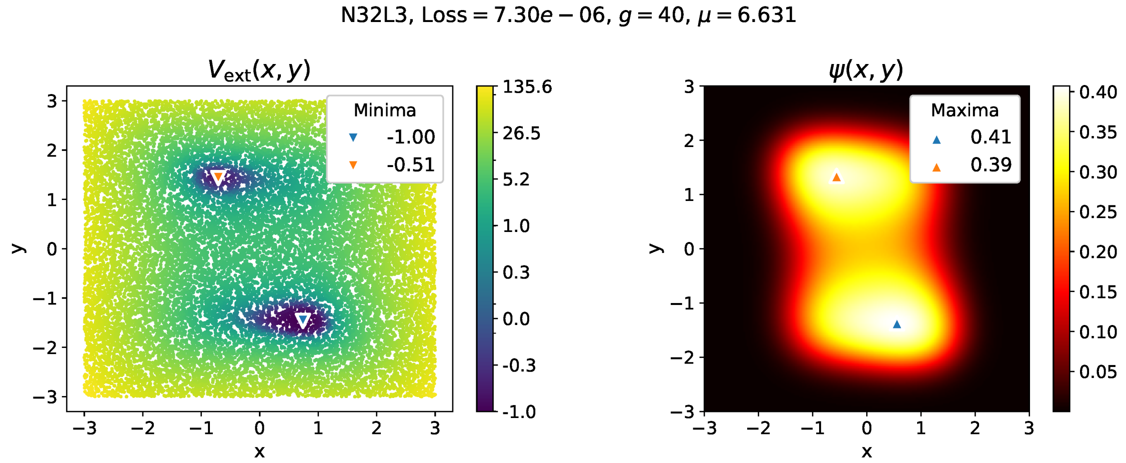

Figure 13: 2D GPE with the asymmetric double well potential from equation (7) and $g = 40$, using a NN with N32L3. At high interaction strength the wave function easily spreads over both wells.

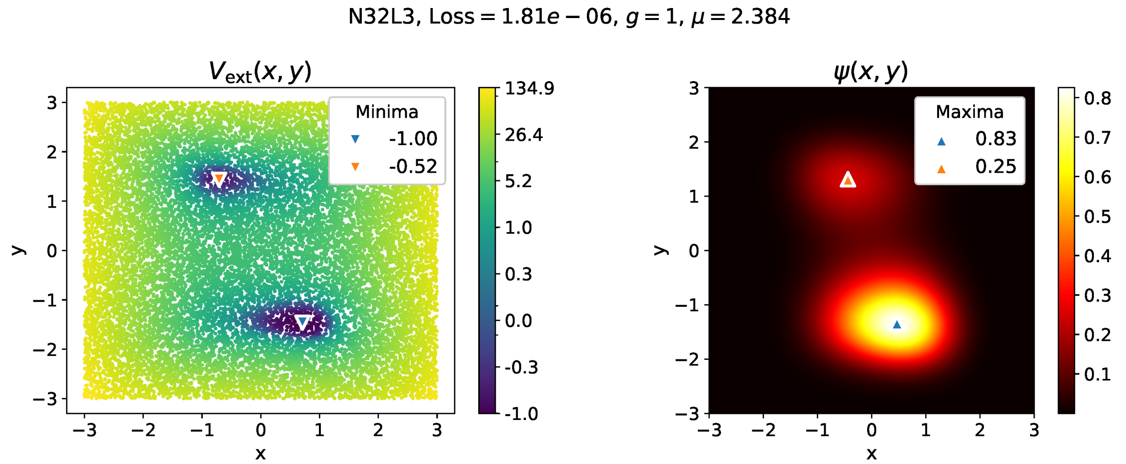

Figure 14: The NN of Figure 13 retrained for $g = 1$. The wave function is now more localized in the lower well of the asymmetric potential, as expected.

3.3 A Modulated Mexican Hat

Further testing the robustness of the PINN approach, we consider a highly structured two-dimensional trap obtained by combining a radial "Mexican-hat" confinement with an angular modulation,

$$V_{\mathrm{ext}}(x, y) = \left(x^2 + y^2 - 2\right)^2 + \mathcal{E}\cos(5\theta), \qquad \theta = \operatorname{atan2}(y, x)\,, \tag{8}$$

where the first term confines the density on a ring of radius $r_0 = \sqrt{2}$, while the second term breaks rotational symmetry and produces five preferred angular minima along the ring. Our goal is to solve the GPE with $g = 1$ at two values of the modulation strength: $\mathcal{E} = 0.4$ and $\mathcal{E} = 10$. Considering the potential landscape we also increased the number of sample points to $N_f = 40000$, the loss term $\mathcal{L}$ remains otherwise the same as in equation (6).

The corresponding solutions are shown in Figures 15 and 16. In both cases the network accurately reproduces the angular density modulations induced by the potential. For $\mathcal{E} = 0.4$ the trap is only weakly corrugated, and the training procedure was quite straightforward, it was sufficient to solve the problem with $g = 40$ and then retrain the network for $g = 1$ to find the final solution. For $\mathcal{E} = 10$ on the other end, the trap is strongly corrugated and the wave function is highly localized in the angular minima, which made the optimization challenging.

To obtain the result in Figure 15, it was inconvenient to find an interaction strength g high enough

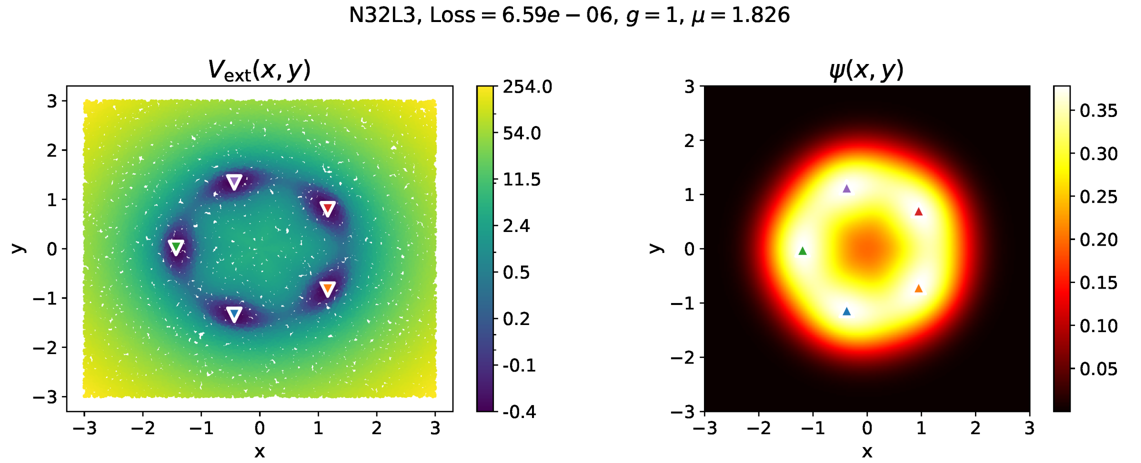

Figure 15: Stationary solution of the 2D GPE with the potential (8) at $\mathcal{E} = 0.4$ and $g = 1$, obtained with a N32L3 PINN. The wave function follows the five-fold modulation of the external trap along the ring. The potential minima are denoted with a ▼ symbol and are all $V_{\text{min}} \simeq 0.40$ deep, while the maxima of the wave-function are denoted with a ▲ symbol and are all $\psi_{\text{max}} \simeq 0.38$ high.

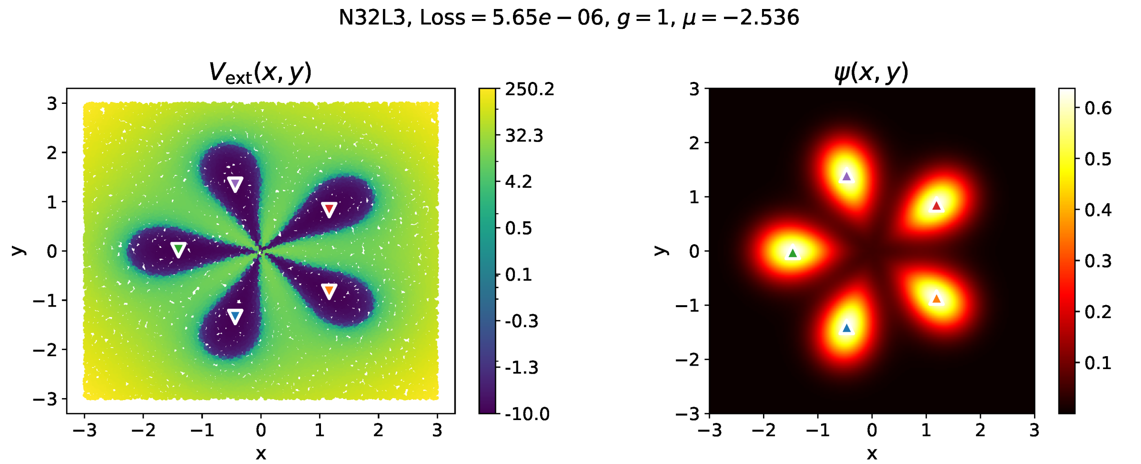

Figure 16: Same as Figure 15, but with a much stronger angular corrugation, $\mathcal{E} = 10$ meaning that the angular barriers are $\Delta V \simeq 20$ high. Even in this stiff regime the PINN converges to a solution that captures the sharp modulation and the increased localization in the angular minima. The potential minima are denoted with a ▼ symbol and are all $V_{\text{min}} \simeq -10$ deep, while the maxima of the wave-function are denoted with a ▲ symbol and are all $\psi_{\text{max}} \simeq 0.64$ high.

to overcome the angular barriers that have a $\Delta V \simeq 20$, consequently we leveraged the situation to see if modifying the potential would work as well. We started from the NN obtained in Figure 15 with $\mathcal{E} = 0.4$ and $g = 1$, and gradually increased $\mathcal{E}$ until we reached the target value of $\mathcal{E} = 10$. At each step we let the optimizer run up to convergence (that was always in the order of 10^{-5}) before increasing $\mathcal{E}$ again. It was crucial to use small increments of $\mathcal{E}$, otherwise we could observe the optimizer getting stuck at a loss of $\sim 10^{-4}$, and the solution dropping a couple of the density peaks entirely. But in the end, thanks to a combination of a warm high g start and a slow modification of the potential we could reach the final solution.

An even more challenging potential with more wells is shown in Appendix B.

4 Vortex Solutions

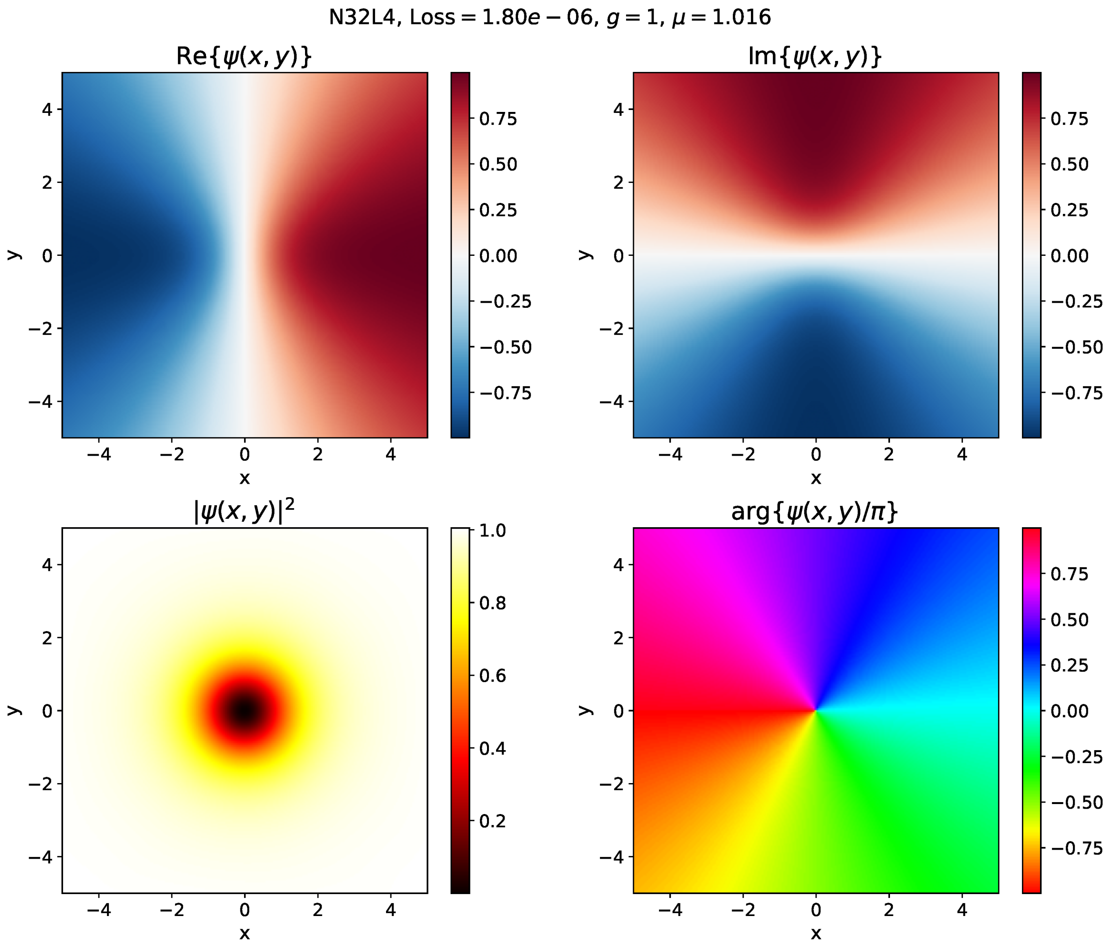

Figure 17: Vortex solution of the 2D GPE with $g = 1$ and winding number $m = 1$, obtained with a NN with N32L4.

As a final application we search for quantized vortex solutions of the stationary GPE in 2D without an external potential,

$$-\frac{1}{2}\nabla^2\psi + \left(g|\psi|^2 - \mu\right)\psi = 0\,. \tag{9}$$

A vortex of winding number m is a solution of the form

$$\psi(r,\theta) = f(r)\,e^{im\theta}\,, \tag{10}$$

where $f(r) \geq 0$ is a real radial profile that vanishes at the origin, forming the vortex core, and saturates to a constant density at large r. Such a state carries a quantized circulation and a phase that winds m times around the core.

We represent the complex wave function with the same two-output NN used in Section 2,

$$\mathcal{N}(x,y) = \left[u(x,y),\ v(x,y)\right], \qquad \psi = u + iv\,.$$

Separating real and imaginary parts of equation (9) and denoting $|\psi|^2 = u^2 + v^2$, we obtain the coupled system

$$f_u \equiv -\frac{1}{2}\nabla^2 u + \left(g|\psi|^2 - \mu\right)u = 0\,, f_v \qquad \equiv -\frac{1}{2}\nabla^2 v + \left(g|\psi|^2 - \mu\right)v = 0\,.$$

Considering that we are looking for an excited state with a specific angular momentum, the boundary conditions play a crucial role as we need to rule out the trivial solution where ψ is a constant. On the boundary $\partial\mathcal{D}$ we directly impose the phase winding from the ansatz (10) with unit amplitude,

$$u(x,y)\big|_{\partial\mathcal{D}} = \cos(m\theta) = \mathrm{Re}\left\{\left(\frac{x+iy}{\sqrt{x^2+y^2}}\right)^m\right\}$$
$$v(x,y)\big|_{\partial\mathcal{D}} = \sin(m\theta) = \mathrm{Im}\left\{\left(\frac{x+iy}{\sqrt{x^2+y^2}}\right)^m\right\}$$

These conditions fix the winding number while leaving the interior amplitude free to relax and impose an asymptotically constant density of $|\psi|^2 \approx 1$. Considering equation (9) when $|\psi|^2$ is constant we have immediately that $\psi = \sqrt{\frac{\mu}{g}}$ for the solution to be consistent, so we expect the chemical potential μ to be close to g for the solution to be correct.

The total loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{PDE}} + \mathcal{L}_{\mathrm{BC}}\,, \tag{11}$$

$$\mathcal{L}_{\mathrm{PDE}} = \frac{1}{N}\sum_{i=1}^{N}\Big[f_u(x_i,y_i)^2 + f_v(x_i,y_i)^2\Big]\,, \tag{12}$$

$$\mathcal{L}_{\mathrm{BC}} = \frac{1}{N_b}\sum_{i=1}^{N_b}\Big[\big(u(x_i,y_i)-\cos(m\theta_i)\big)^2 + \big(v(x_i,y_i)-\sin(m\theta_i)\big)^2\Big]\,, \tag{13}$$

where $N = 20000$ collocation points are sampled from the interior with LHS, $N_b = 400$ points are distributed with LHS along $\partial\mathcal{D}$, and the chemical potential μ is a trainable parameter.

Figure 17 shows the results for a vortex with winding number $m = 1$ and interaction strength $g = 1$ obtained with a NN with N32L4, where the vortex core and the expected phase winding are clearly visible. Increasing the winding number to $m = 3$ results in a larger vortex core, as shown in Figure 18, which is consistent with the fact that the centrifugal barrier increases with m.

Finally, increasing the interaction strength to $g = 10$ makes the vortex smaller. So we could find a solution with a higher winding number $m = 6$ without enlarging the domain, as shown in Figure 19.

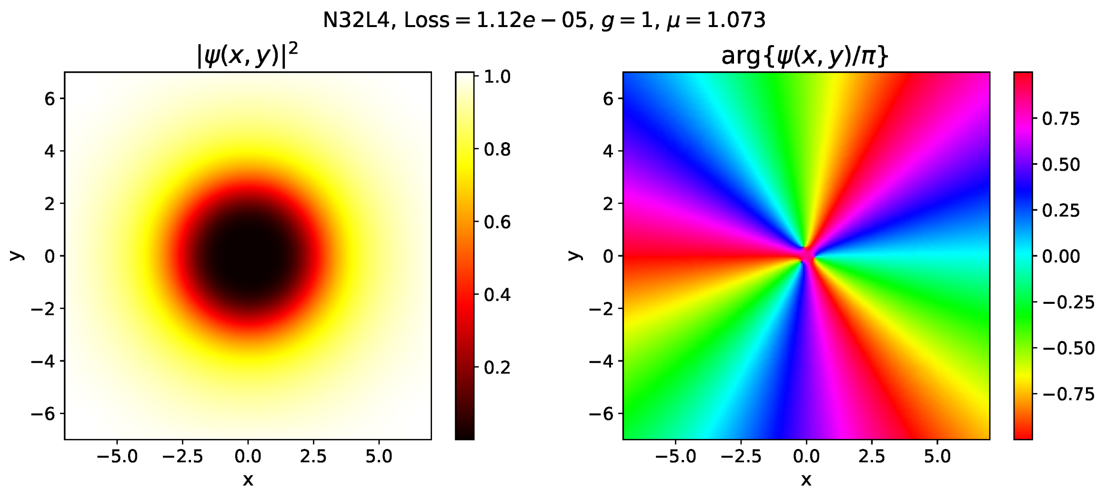

Figure 18: Vortex solution of the 2D GPE with $g = 1$ and winding number $m = 3$, the vortex is bigger than the one with $m = 1$.

Conclusion

In this report we have demonstrated the effectiveness of Physics-Informed Neural Networks as a flexible and accurate tool for solving nonlinear quantum mechanical problems governed by the Gross-

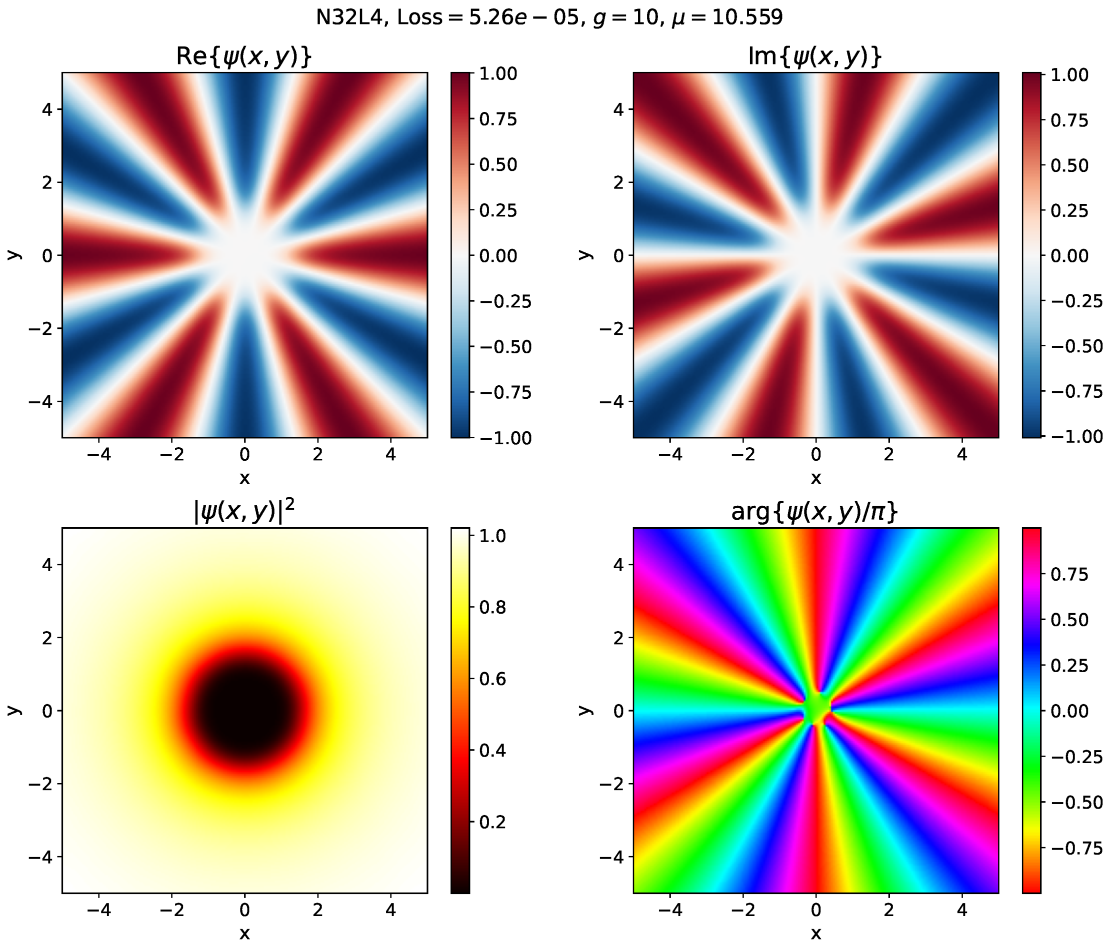

Figure 19: Vortex solution of the 2D GPE with $g = 10$. When the interaction is stronger the vortex is smaller, allowing to find a solution with a higher winding number $m = 6$ in a small domain.

Pitaevskii equation, in both one and two spatial dimensions.

Starting from the 1D radial GPE with a harmonic potential, we showed that even very compact architectures (as small as N4L1, with only 13 trainable parameters) are capable of reproducing the ground state wave function with relative $\mathbb{L}_2$ errors of order 10^{-4}–10^{-5}. Notably, the PINN approach proved particularly valuable for the strongly interacting case $Na = 100$, where conventional methods showed convergence issues, and the NN unambiguously confirmed the variational result. Moving to the time-dependent nonlinear Schrödinger equation in 2D, we reproduced the benchmark of Raissi et al. Raissi, Perdikaris, and Karniadakis 2019 using a significantly smaller network (N32L3 vs. N100L5), confirming that the framework is competitive even with reduced computational resources.

The double well problems highlighted the main challenge of PINNs for eigenvalue problems: the tendency to concentrate the wave function in a single well and neglect the rest once the normalization constraint is satisfied. We addressed this with a continuation strategy, initializing the training at high interaction strength g and progressively reducing it to the target value, which proved effective not only in 1D and 2D double wells, but also for a more complex 2D trap like a modulated Mexican hat. As a final application, by directly encoding the phase winding as a boundary condition, the PINN spontaneously developed quantized vortex solutions with winding numbers up to $m = 6$, correctly reproducing the expected dependence of the core size on both m and g.

Across all problems where independent, if not exact, data was available, a low training loss consistently correlated with a low $\mathbb{L}_2$ error, validating it as a reliable quality metric. The choice of architecture, sampling points and loss weights remains more of an art than a science, yet once the right constraints are in place the network converges to physically meaningful solutions, and in several cases proved more robust than conventional approaches. What makes PINNs particularly appealing is a certain *naturalness* as a computational tool for physics: the absence of a mesh, automatic differentiation, and the ability to encode physical laws directly into the loss function make the method feel closer to the physics than a finite-difference discretisation, with the added practical advantage of being able to resume and continue the training of an already converged network.

A The Eigenvalue Problem

The results obtained in section 3.1 looked solid, but we might still be worried that the procedure of starting from a highly interacting regime, increasing the nonlinearity of the PDE, might have found some incorrect solutions. Our NN could, for example, be in an excited metastable state.

In this section we present a method to study the spectrum of the system and find eigenvalues and eigenfunctions of the GPE in a more self-consistent way. The only caveat is that the spectral theorem we rely on is only valid for linear operators and the GPE is nonlinear.

We start by considering that the eigenstates, solutions of a Schrödinger equation, are orthogonal to each other, so we can enforce this constraint when training the NN. In particular, we can rewrite the problem as

$$-\frac{1}{2}\frac{\mathrm{d}^2\psi_n(x)}{\mathrm{d}x^2} + \left(V_{\text{ext}}(x) + g|\psi_n(x)|^2 - \mu_n\right)\psi_n(x) = 0 \tag{14}$$

$$\int_{x_l}^{x_r} \psi_n(x)\psi_m(x)dx = 0, \quad n \neq m \tag{15}$$

and define a NN that takes n as input too $\mathcal{N}(x,n)$, so that the loss function becomes

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}} + 10\ \mathcal{L}_{\text{norm}} + \mathcal{L}_{\text{ortho}} \tag{16}$$

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_{\text{s}}N}\sum_{n=1}^{N_{\text{s}}}\sum_{i=1}^{N}\left(-\frac{1}{2}\frac{\mathrm{d}^2\psi_n(x_i)}{\mathrm{d}x^2} + \left(V_{\text{ext}}(x_i) + g|\psi_n(x_i)|^2 - \mu_n\right)\psi_n(x_i)\right)^2 \tag{17}$$

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_{\text{s}}}\sum_{n=1}^{N_{\text{s}}}\frac{|\psi_n(x_l)|^2 + |\psi_n(x_r)|^2}{2} \tag{18}$$

$$\mathcal{L}_{\text{norm}} = \frac{1}{N_{\text{s}}}\sum_{n=1}^{N_{\text{s}}}\left(\frac{x_r - x_l}{N}\sum_{i=1}^{N}\psi_n(x_i)^2 - 1\right)^2 \tag{19}$$

$$\mathcal{L}_{\text{ortho}} = \sum_{n\neq m}\left(\frac{x_r - x_l}{N}\sum_{i=1}^{N}\psi_n(x_i)\psi_m(x_i)\right)^2. \tag{20}$$

We can now train the NN on a grid of $x \in [x_l, x_r]$ and $n \in [0,\ 1,\ \ldots,\ N_{\text{s}} - 1]$ where each n corresponds to a different eigenstate and orthogonality is enforced between all the states. Of course, we have no guarantee that the NN will learn the states in the correct order, nor that it will learn the lowest energy ones, but if we choose $N_{\text{s}} = 6$ we still have a good chance of finding the ground state.

Figure 20 shows the results for the harmonic potential with $g = 0$, of which the spectrum is known analytically to be $\mu = \frac{1}{2} + \tilde{n},\ \tilde{n} = 0, 1, \ldots$, and we can see that the NN correctly learns the 6 eigenstates of the system (that we ordered from the one with the lowest μ to the one with the highest μ), but happened to miss the 5th one.

Figures 21 and 22 show instead the results for the symmetric and non-symmetric double well potentials used in section 3.1 (Figures 10 and 11 respectively).

This method is not intended to find the entire spectrum of the system, nor to accurately predict the wave functions, but it has a high chance of finding the ground state immediately and in a self-consistent way, considering that every state found must be orthogonal to the others. We can also use the results to get an idea of the spectrum of the system and use this information to guide the optimizer in the early stages of the training that was done in section 3.1.

A more precise assessment of the precision of the ground state found with this method is presented in Figure 23 and Figure 24, where we compare the results obtained with the two different methods. A non-perfect agreement is to be expected considering that when computing more eigenstates the optimizer only reached a loss of $\sim 10^{-4}$, while in the previous section we were able to reach a loss of $\sim 10^{-6}$. For the double well potential with $a = 2$ the relative $\mathbb{L}_2$ error is 9.92×10^{-3}, while for the non-symmetric double well potential it is 4.43×10^{-3}. Part of the error could also be due to the fact that imposing the orthogonality constraint would only be justified when $g = 0$ (when the GPE reduces to a linear Schrödinger equation), while in this case we have $g = 1$. Indeed, even at $g = 0$ it can be a starting point to find the solution of the interacting problem.

Extending this method to 2D for a harmonic potential we correctly recover the ground states and the excited eigenstates with non-zero angular momentum, as shown in Figure 25.

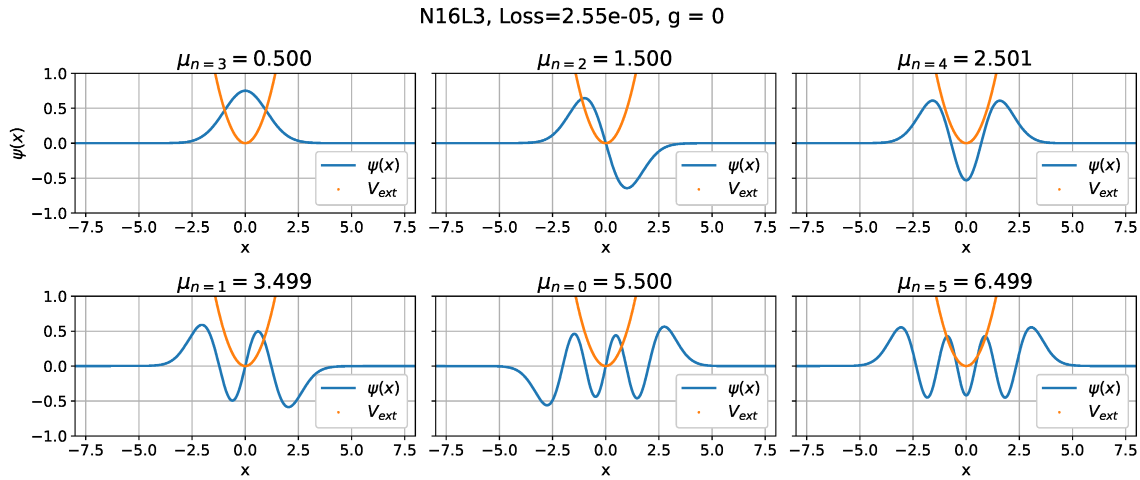

Figure 20: 1D GPE with $g = 0$ and a harmonic potential. The same NN correctly learns 6 eigenstates (that in this case are known analytically) but misses the 5th one.

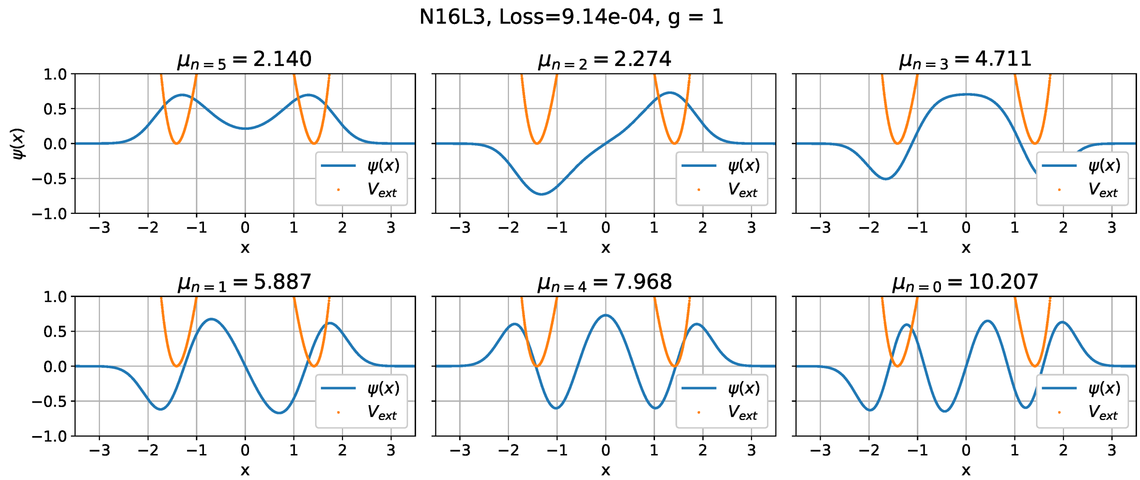

Figure 21: Results for the 1D GPE with $g = 1$ and a symmetric double well potential from equation (4) with $a = 2$ using a NN with N16L3.

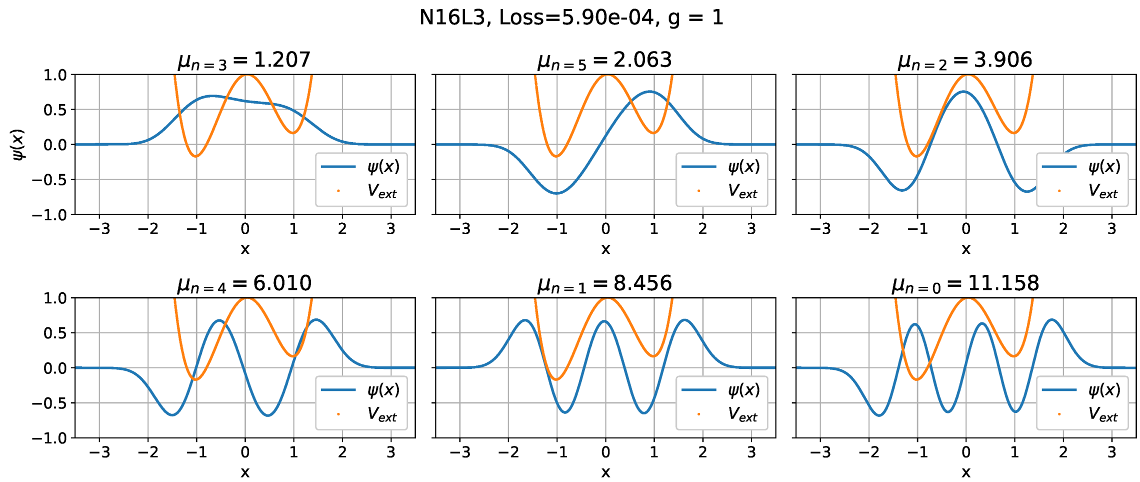

Figure 22: Results for the 1D GPE with $g = 1$ and a non-symmetric double well potential from equation (5) with $a = 1$ and $b = 6$ using a NN with N16L3.

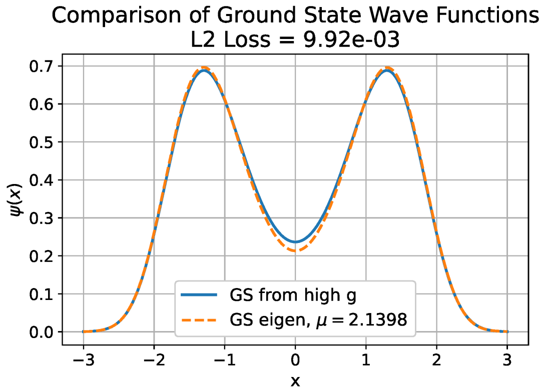

Figure 23: Comparison of the results obtained for the double well potential with $a = 2$ and $g = 1$ with the method of section 3.1 (solid blue line) and the method of Appendix A (orange dotted line).

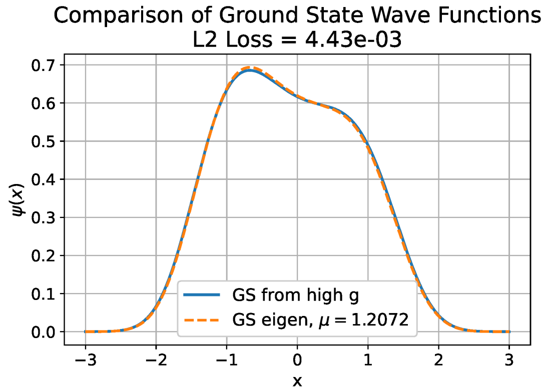

Figure 24: Comparison of the results obtained for the non-symmetric double well potential with $a = 1$, $b = 6$ and $g = 1$ with the method of section 3.1 (solid blue line) and the method of Appendix A (orange dotted line).

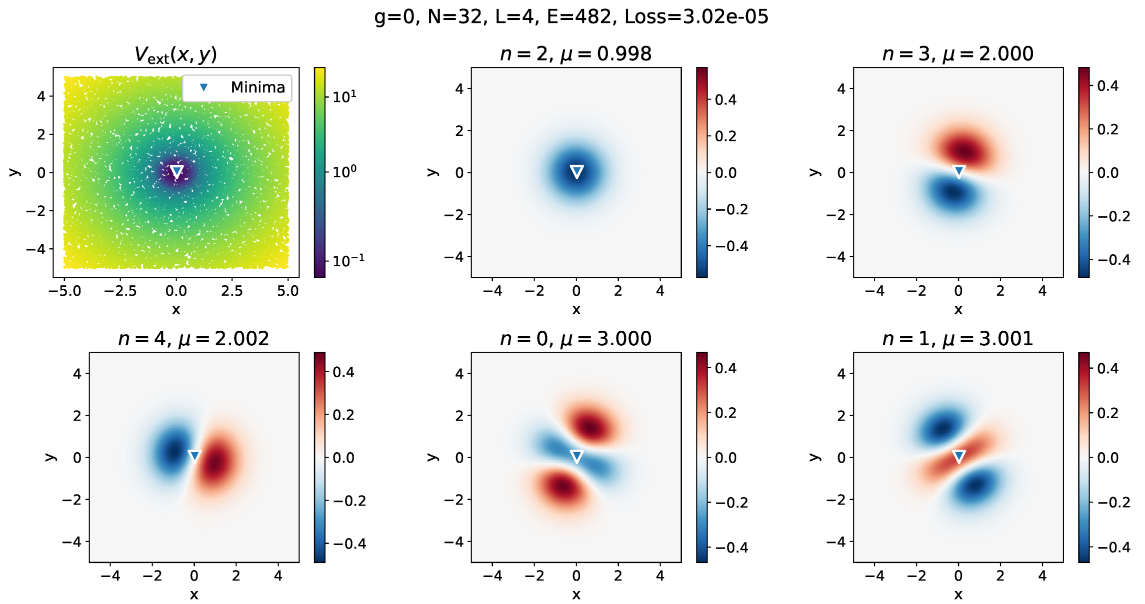

Figure 25: Five eigenstates of the 2D GPE with $g = 1$ and a harmonic potential, computed with a N32L4 NN. The ground state ($n = 3$) and two pairs of degenerate excited states are found, with the degeneracy consistent with the rotational symmetry of the harmonic potential.

For a double well however, the method fails, as the NN consistently converges to an excited state with the probability localized in a single well and never to the ground one.

B Spread-Out Potential

In the section 3.1, we developed a strategy to force the wave function to spread over the entire domain by increasing the interaction strength g. Later we observed that this method worked well in 2D (Section 3.2) and with a more complex potential (Section 3.3).

As a further test, we present here an even more complex potential to test the ability of the PINN to spread the wave function over multiple wells distributed across the domain.

$$V_{\text{ext}}(x, y) = \frac{1}{2}\,\omega(x^2 + y^2) + V_0 \sum_{k=0}^{4} \cos\left(2\pi\left(x\cos\theta_k + y\sin\theta_k\right)\right), \qquad \theta_k = \frac{2\pi k}{5}\,, \tag{21}$$

is a superposition of a harmonic trap and a five-fold angular modulation that creates a pattern of small potential wells, a representation of which is shown in the left panel of Figure 26.

We use $\omega = 0.1$ and $V_0 = 1$, so that the potential is not dominated by the harmonic term and the wave function is expected to spread more over the entire domain (while still being confident by the harmonic trap). For this reason, we also drop the boundary condition requiring the wave function to vanish at the boundary. The loss therefore consists only of the PDE term $\mathcal{L}_{\text{PDE}}$ and the normalization term $\mathcal{L}_{\text{norm}}$.

Figure 26 shows the potential alongside the solution obtained with an N32L3 neural network for $g = 1000$. We use 40,000 sample points to compute $\mathcal{L}_{\text{PDE}}$ and $\mathcal{L}_{\text{norm}}$. Inspecting the potential (left panel of Figure 26) we observe an inner ring with a minimum of -1.94, a ring of small wells with a common minimum of -3.11, and an outer ring of even smaller wells with a common minimum of -1.56.

We then proceed by reducing g by 10% at each step and retraining the network down to $g = 1$, each time reaching `L-BFGS` convergence with a loss below 10^{-4}. Two intermediate solutions are shown in Figure 27 to appreciate how the wave function becomes progressively more localized, but still keeping the pattern of the potential in the outer ring of wells. Finally, the solution for $g = 1$ is shown in Figure 28, where we can see that the wave function is mostly localized in the inner ring and in the middle ring, even though the wells in the middle ring are deeper. On the right panel of Figure 28 we stretch the color scale to better visualize that the effect of the outer ring is still somewhat visible, even though the wave function is very small in that region.

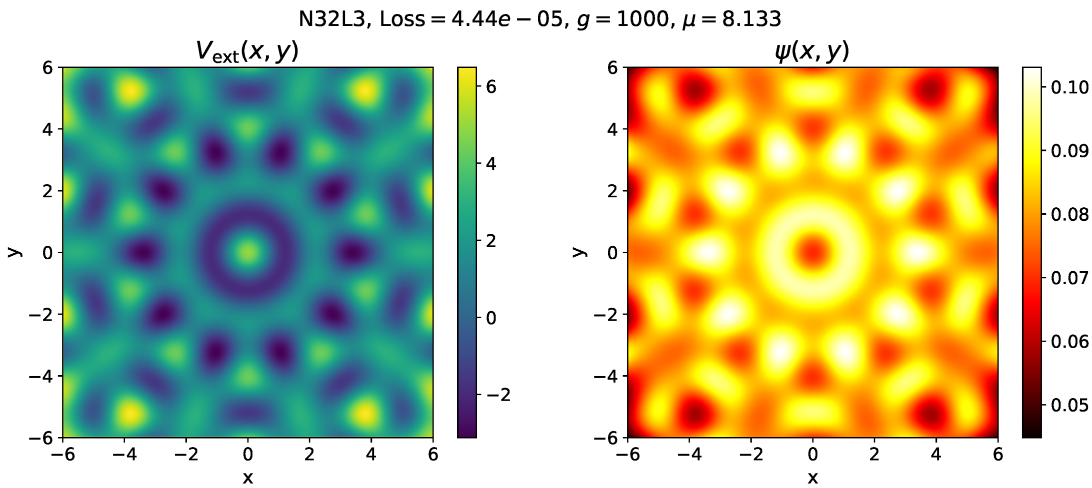

Figure 26: External potential $V_{\text{ext}}(x, y)$ defined in (21) (left) and corresponding ground-state solution $\psi(x, y)$ (right) obtained with an N32L3 network for $g = 1000$. The strong interaction leads to a wave function that spreads over multiple rings of local minima.

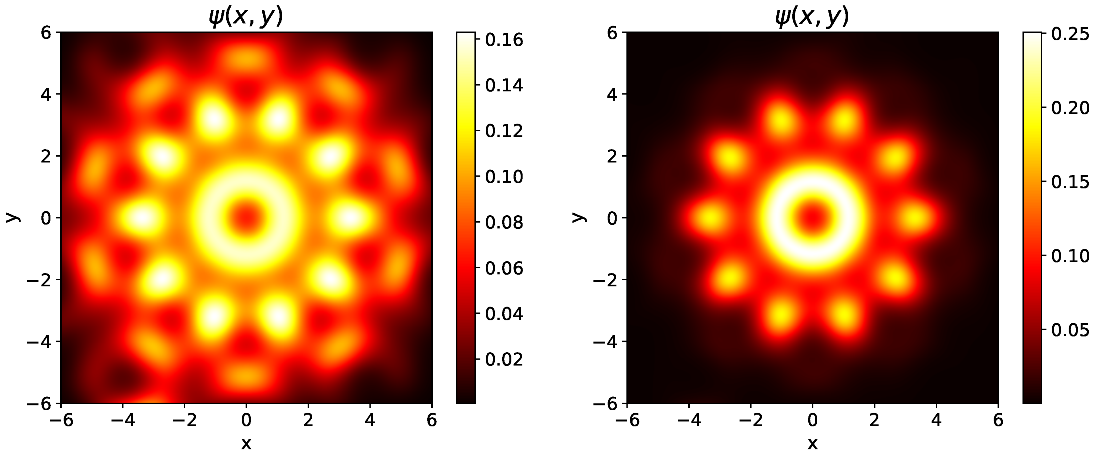

Figure 27: Ground-state solutions $\psi(x, y)$ obtained during the continuation procedure as g is reduced. Left: $g = 106.6$. Right: $g = 10.5$. As the interaction strength decreases, the solution becomes progressively more localized around the energetically favorable rings.

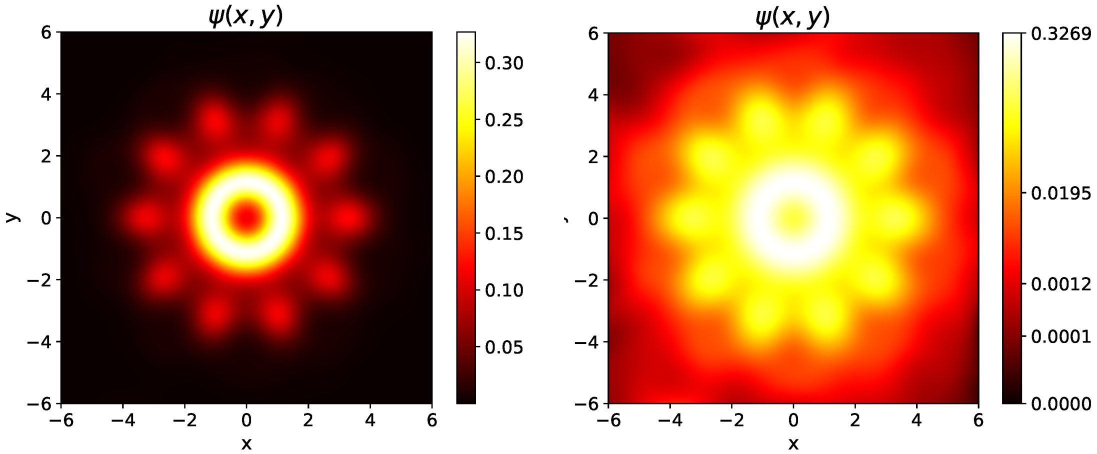

Figure 28: Ground-state solution for $g = 1$ computed with additional sampling points to improve resolution. On the right we stretch the color scale to better visualize the value of the wave function close to 0 and observe that the solution still conserves the shape of the potential in the outer ring of wells. For weak interactions, the wave function concentrates primarily in the innermost ring, even though the middle wells are deeper.

References

Raissi, M., P. Perdikaris, and G.E. Karniadakis (2019). "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations". In: *Journal of Computational Physics* 378, pp. 686–707. ISSN: 0021-9991. DOI: `https://doi.org/10.1016/j.jcp.2018.10.045`. URL: `https://www.sciencedirect.com/science/article/pii/S0021999118307125`.